

Gear Train Design

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Overall Design Process:

From the first iteration was guessing a reasonable input and output RPM along with a fitted HP that a real-life motor can provide. Next was calculating the number of gear teeth my input and output RPM ratio would require and check to see if any adjustments to the RPMS were needed. Through the basic specs from this information, I chose a gear from the catalog to serve as an initial test for future calculations and iterations. Then I identified suitable motor, keyway and retaining rings that can be physically obtained.

The next step was calculating the forces and torques acting on the intermediate shaft to calculate to max moment. My initial max forces and moments were too extreme so I decided to go forward with larger gears that would allow for a bigger bore diameter. After this adjustment, I calculated the possible bearings I would use to get the full estimated length of my shaft.

I then calculated the stress at three points where I believed the critical stress would occur. These calculations highlighted that my first estimated diameter was too small. Through iteration I calculated a larger diameter providing the FOS of around 2.5 in the keyway. Once my deflection was properly calculated, I went through another iteration to achieve the final diameter of my shaft. Then I fully optimized my shaft length ensuring all pieces could fit and were in tolerance and created my model in SOLIDWORKS to double check my calculations with and FEA. This was the design process I underwent to design the intermediate shaft.

Input and Output RPM/HP:

- Input RPM: 1750
- Input hp: 15
- Desired output RPM: 85

Gear Ratios: (Compound reverted gear train)

$$e = \frac{\text{Output RPM}}{\text{Input RPM}} = \frac{85 \text{ RPM}}{1750 \text{ RPM}} = 0.04857$$

$$e = \frac{1}{20.59} = \frac{N_2 N_4}{N_3 N_5}$$

Identical Reductions:

Utilizing identical reductions for use of compound reverted gear train to get a compact gear box

$$\frac{N_2}{N_3} = \frac{N_4}{N_5}$$

$$e = \left(\frac{N_2}{N_3}\right)^2 \Rightarrow \frac{N_2}{N_3} = \sqrt{e}$$

$$\frac{N_2}{N_3} = \frac{1}{4.54} = \frac{N_4}{N_5}$$

Assume pressure angle $\phi = 20$ and $k = 1$

Calculating number of teeth required on the gears.

$$m = 4.54$$

$$N_p = \frac{2k}{(1 + 2m)\sin^2(j)} * (m + \sqrt{m^2 + (1 + 2m)\sin^2(j)})$$

$$N_p = \frac{2(1)}{(1 + 2 * 4.54)\sin^2(20)} * (9 + \sqrt{9^2 + (1 + 2 * 4.54)\sin^2(20)})$$

$$N_p = 15.6 = 16$$

$$N_2 = N_4 = 16 \text{ Teeth}$$

$$N_3 = N_5 = N_p * m = 16 * 4.54 = 72.64 = 72 \text{ Teeth}$$

$$\text{Output RPM } (w) = \frac{16}{72} * \frac{16}{72} * 1750 = 86.42 \text{ RPM vs } 85 \text{ RPM (acceptable)}$$

Loads and Torque Calculations:

$$\omega_3 = \omega_2 * e = 1750 * \frac{16}{72} = 388.9 \frac{\text{rev}}{\text{min}} p = T_2 \omega_2 = T_3 \omega_3 = T_4 \omega_4 = T_5 \omega_5$$

$$T_2 = \frac{Hp}{\omega_2} = \frac{15 \text{ Hp}}{1750 \text{ RPM}} * \frac{550 \frac{\text{ft} \cdot \text{lb}}{\text{s}}}{1 \text{ Hp}} * \frac{1 \text{ rev}}{2\pi \text{ rad}} * \frac{60 \text{ s}}{\text{min}} = 45 \text{ lbf} \cdot \text{ft}$$

$$T_2 = 45 \text{ lbf} * ft$$

$$T_3 = T_2 * \frac{\omega_2}{\omega_3} = 45 * \frac{1750}{388.9} = 202 \text{ lbf} * ft = T_4$$

$$T_3 = T_4 = 202 \text{ lbf} * ft$$

$$T_5 = T_2 * \frac{\omega_2}{\omega_5} = 45 * \frac{1750}{86.42} = 911.2 \text{ lbf} * ft$$

$$T_5 = 911.2 \text{ lbf} * ft$$

$$\omega_t = \frac{T * 2}{d}$$

$$\omega_{t2} = \frac{T_2}{r_2} = \frac{45}{\frac{2}{12}} = 270 \text{ lbf}$$

$$\omega_{t3} = \frac{T_3}{r_3} = \frac{202}{\frac{9}{12}} = 270 \text{ lbf}$$

$$\omega_{t5} = \omega_{t4} = \frac{T_5}{r_5} = \frac{911.2}{\frac{9}{12}} = 1214.9 \text{ lbf}$$

$$\omega_{n3} = \omega_{n2} = 270 * \tan(20) = 98.27 \text{ lbf}$$

$$\omega_{n4} = \omega_{n5} = 1214.9 * \tan(20) = 442.18 \text{ lbf}$$

Calculating max moment in both y and z planes to find total max moment.

$$M_{max,y} = 1180.53 \text{ lb in} @ x = 7.65$$

$$M_{max,z} = -2600.85 \text{ lb in} @ x = 7.65$$

$$M_{max} = \sqrt{M_{max,y}^2 + M_{max,z}^2} = \sqrt{3299.94^2 + 1201.06^2} = 3511.7 \text{ lb in}$$

Bore max & mins: Gear 16T = 1-1.125 Gear 68T = 1.25-3.25

Cylindrical Roller Bearings:

Calculating the C₁₀ rating to find suitable bearings.

$$C_{10} = F_D \cdot \left(\frac{\mathcal{L}_D n_D 60}{10^6} \right)^{\frac{1}{3}}$$

$$a = \frac{10}{3} \quad \mathcal{L}_D = 10,000 \quad n_D = \frac{1750}{60} = 29.167$$

$$C_{10} (1) = 619.6468 \cdot \left(\frac{10,000 * 29.167 * 60}{10^6} \right)^{\frac{3}{10}} * 0.004448 = 14.1107$$

$$C_{10} (2) = 960.5626 \cdot \left(\frac{10,000 * 29.167 * 60}{10^6} \right)^{\frac{3}{10}} * 0.004448 = 21.8741$$

$$C_{10} \Rightarrow 28.6, \text{Bearing bore} = 25\text{mm} = 0.984252"$$

Chose cylindrical roller bearing chosen for better support based off C_{10} table.

Shaft Length: 13.4" Gear Pos 1: 3.65" Gear Pos 2 = 8.95 Retaining ring thickness = .106"

Calculated an estimated shaft length based on bearing thickness, gear hub + face width, retaining ring thickness, and shoulder.

Code to graph the xy, xz, and combined moment graphs by utilizing singularity functions:

```
x=0:0.0001:11.062;
R1y = 582.29;
R2y = 902.61;
R1z = 211.933;
R2z = 328.517;
G1t = 98.27;
G2t = 442.18;
G1n = 270;
G2n = 1214.9;
x12 = 3.656;
x23 = 7.406;
```

```

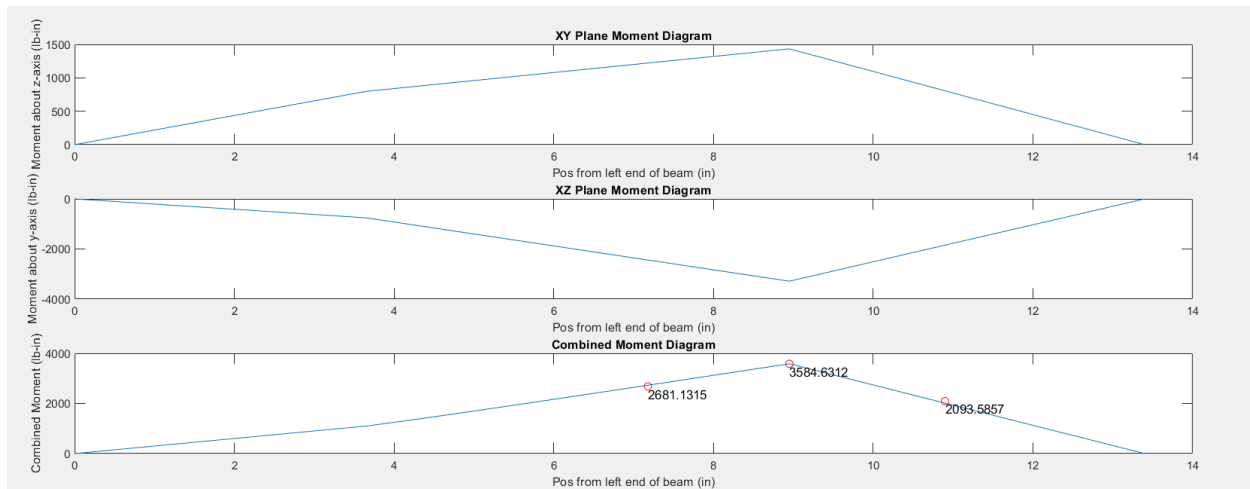
L = 11.062;
test = R1y + R2y - G2n - G1n

Vy = R1y*sin(0,x) - G1n*sin(0,x-x12) - G2n*sin(0,x-x23) + R2y*sin(0,x-L);
Vz = R1z*sin(0,x) - G1t*sin(0,x-x12) - G2t*sin(0,x-x23) + R2z*sin(0,x-L);
Mz = R1y*sin(1,x) - G1n*sin(1,x-x12) - G2n*sin(1,x-x23) + R2y*sin(1,x-L);
My = R1z*sin(1,x) - G1t*sin(1,x-x12) - G2t*sin(1,x-x23) + R2z*sin(1,x-L);

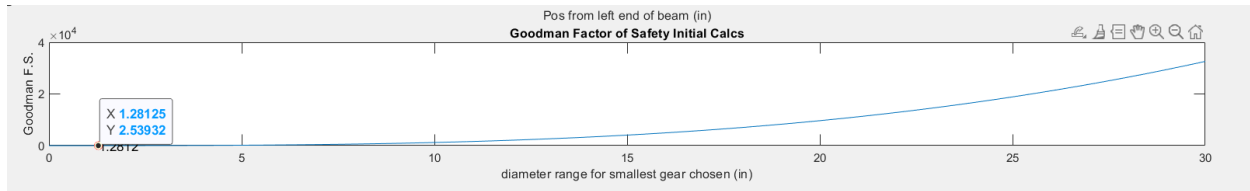
Mtot = sqrt(My.^2 + Mz.^2);
%shear in the y
subplot(2,2,1)
plot(x, Vy)
title("Vy")
%shear in the z
subplot(2,2,2)
plot(x, Vz)
title("Vz")
subplot(2,2,3)
plot(x, Mz)
title("Mz")
subplot(2,2,4)
plot(x, My)
title("My")

figure(2)
plot(x, Mtot)
title("Mtot")

```



(Produced by shaft_iterate)



Goodman FOS comparison:

4142 Q & T & Deformed steel properties chosen for its high ultimate tensile strength

Dmin = 1.28125

Se = 200 kpsi Sut = 295 kpsi

$$n_f = \frac{\pi d^3}{16} * \left(\frac{\sigma'_a}{S_e} + \frac{\sigma'_m}{S_{ut}} \right)^{-1}$$

$$q = \frac{(k_f - 1)}{(k_t - 1)}$$

Code to calculate the von mises stress and FOS in the keyway.

Testing to check if estimated diameter 1.28125 is withing tolerance for my 2.5 FOS within critical stress points.

Key points: 3584 lb-in Keyway @ 8.95

```
d = 1.28125;
Sut = 207;
Se = 200;
TorG3 = 202;
q = 0.865;
```

```
%keyway
```

```
Mmax = 3584;
Mmin = -3584;
ktB = 2.1;
ktsT = 3;
kfB = 0.95*(ktB-1) + 1;
kfsT = 0.95*(ktsT-1) + 1;
sigma_bend = ((Mmax-Mmin)/2 *(d/2))/((pi*d^4)/64);
tao_tor = (TorG3*(d/2))/((pi*d^4)/32);
von_amp = ((kfB)*sigma_bend)/1000;
von_mean = 3*(kfsT*tao_tor)/1000;

keyway_nf = ((pi*d^3)/16)*(von_amp/Se + von_mean/Sut)^(-1)
```

Code to calculate the goodman FOS in the shoulder.

Key Points 2681 lb-in Shoulder @ 7.1793

```
%shoulder
Mmax1 = 2681;
Mmin1 = -2681;
ktB1 = 2.4;
ktsT1 = 2.0;
kfB1 = 0.95*(ktB-1) + 1;
kfsT1 = 0.95*(ktsT-1) + 1;
sigma_bend1 = ((Mmax-Mmin)/2 *(d/2))/((pi*d^4)/64);
tao_tor1 = (TorG3*(d/2))/((pi*d^4)/32);
von_amp1 = ((kfB)*sigma_bend)/1000;
von_mean1 = 3*(kfsT*tao_tor)/1000;

shoulder_nf = ((pi*d^3)/16)* (von_amp/Se + von_mean/Sut)^(-1)
```

Code to calculate the von mises stress in the retaining ring.

Key Points 2093 lb-in Retaining Ring @ 10.9

t = 2 - 1.88 = 0.12 a = 0.139, a/t = 0.863

```
%ret Groove
Mmax2 = 2093;
Mmin2 = -2093;
ktB2 = 5;
ktsT2 = 1.9;
kfB2 = 0.95*(ktB-1) + 1;
kfsT2 = 0.95*(ktsT-1) + 1;
sigma_bend2 = ((Mmax-Mmin)/2 *(d/2))/((pi*d^4)/64);
tao_tor2 = (TorG3*(d/2))/((pi*d^4)/32);
von_amp2 = ((kfB)*sigma_bend)/1000;
von_mean2 = 3*(kfsT*tao_tor);

ret_nf = ((pi*d^3)/16)* (von_amp/Se)^(-1)
```

Critical Stress at Retaining Ring Groove with new diameter: 1.409"

Solved for deflection using singularity functions given with the shaft_iterate code and is not within tolerance.

Calculating new diameter after deflection:

$$d_{new} = d_{old} \left| \frac{n_d y_{old}}{y_{all}} \right|^{\frac{1}{4}}$$
$$1.6905 = 1.4039 * \left| \frac{(0.002798^2 + 0.00553519^2)^{\frac{1}{2}}}{0.003} \right|^{\frac{1}{4}}$$

Calculating new FOS with the 3rd iteration diameter.

After 3rd Iteration:

ret_nf = 9.433

shoulder_nf = 14.83

Keyway_nf = 11.72

Found using code given above.

Gear 3 New Ideal Diameter: 1.6905"

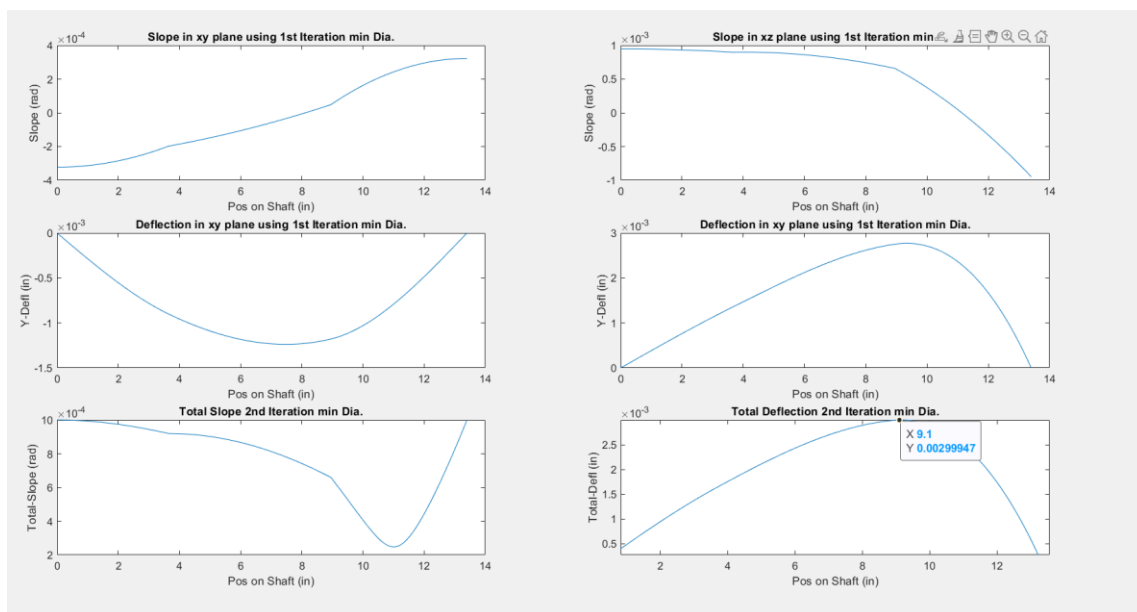
In between Diameter: $1.6905 \times 1.5 = 2.53575$ "

Gear 2 Diameter: 2"

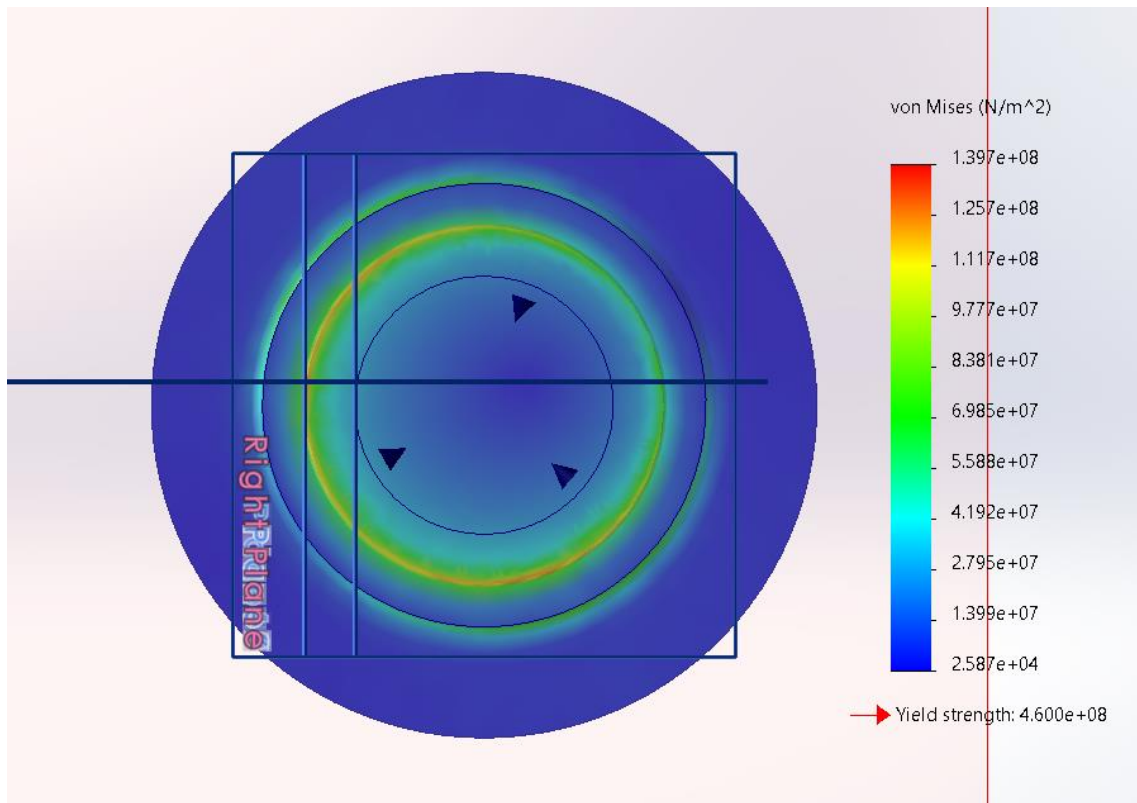
New Keyway sizes: 1 3/8 – 1 1/4, w: 3/8 h: 1/4 depth: 1/8

Solve for slope and max deflection using new diameter to be within criteria.

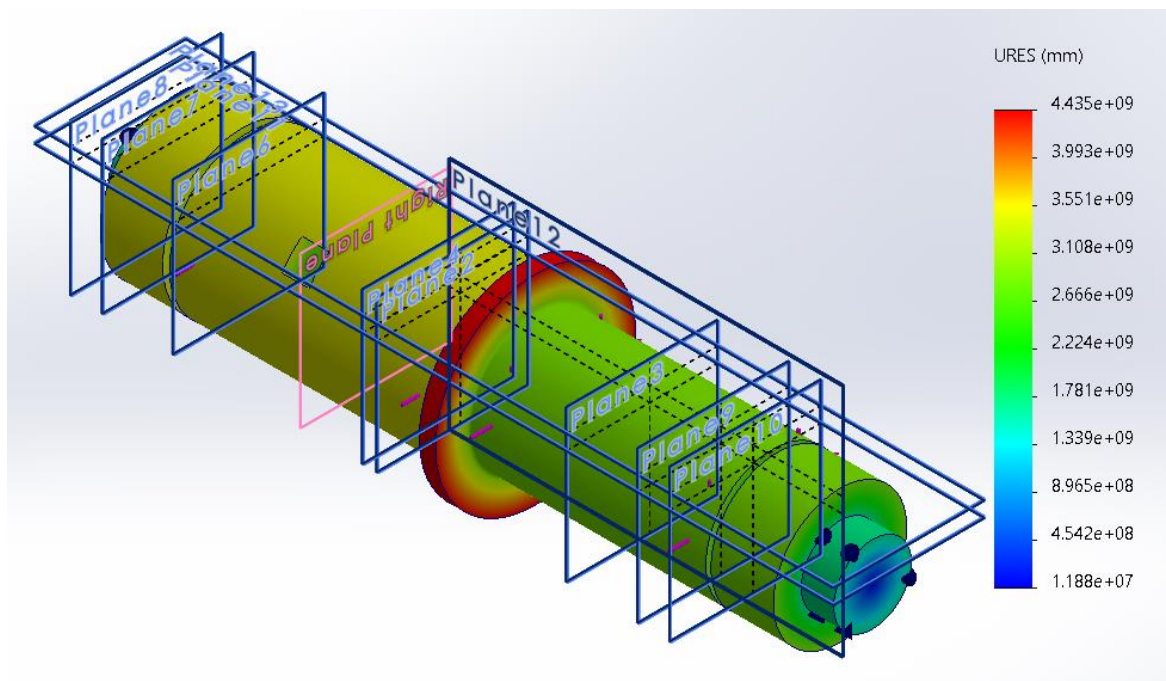
Max Deflection with new diameter: 0.002999



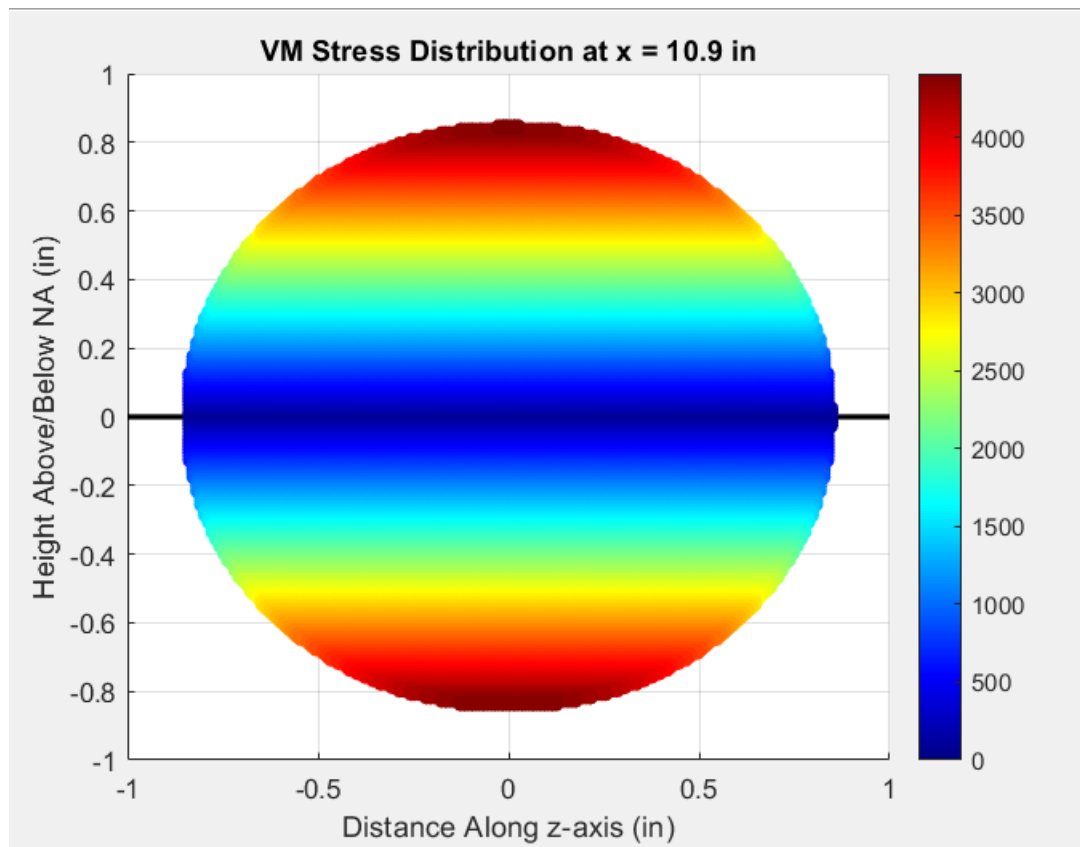
Finite Element Analysis on Solidorks



Deflection analysis on SolidWorks



MATLAB Plot



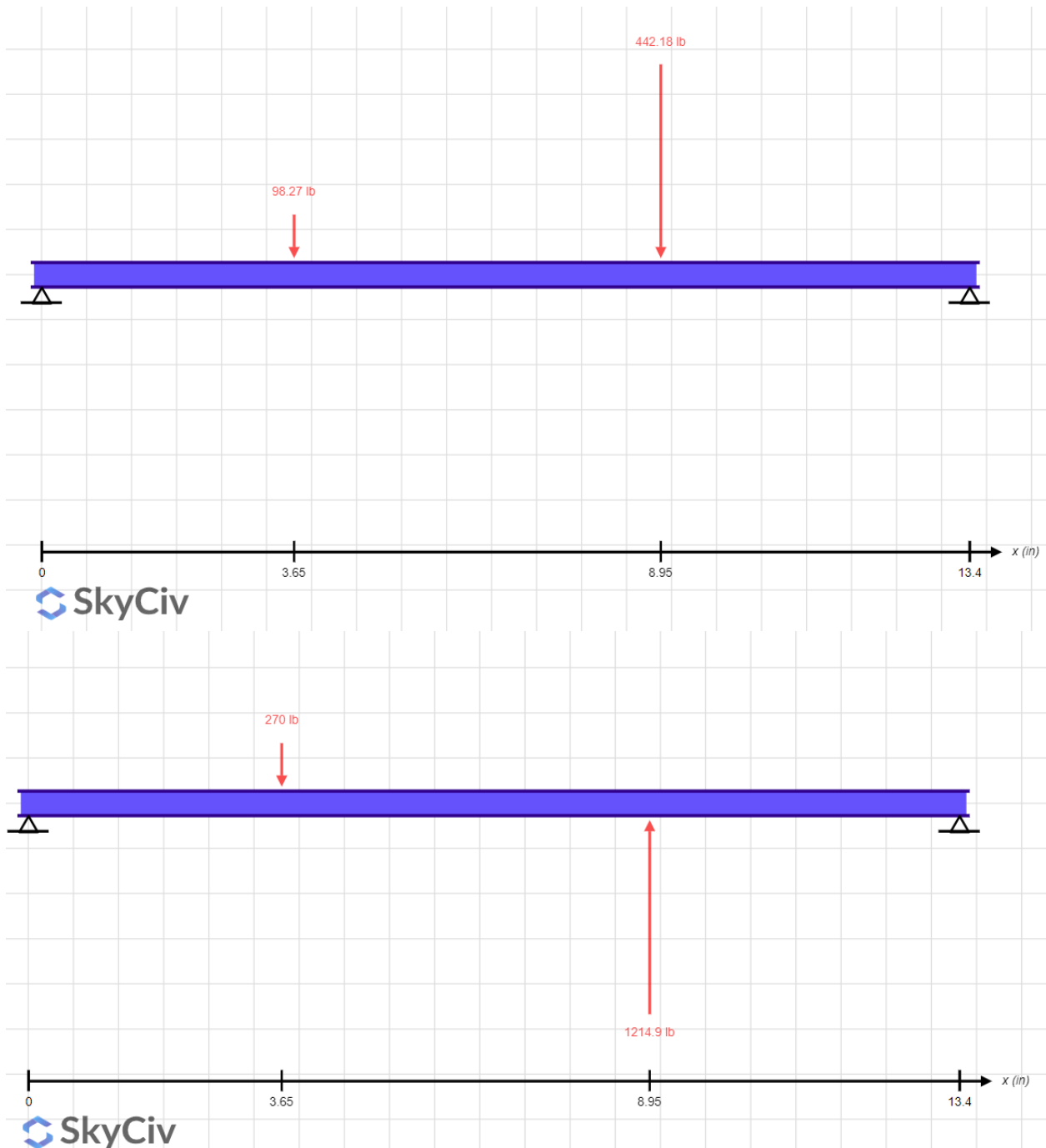
Utilizing combined moment on critical stress point to find the VM stress distribution

```
clear;
clc;
close all;
d = 1.6905;
n = 1;

for z = -0.84525:0.01:0.84525 % what needs to be changed for z and y???
    for y = -0.84525:0.01:0.84525
        theta = atand(y./z);
        rho = sqrt(z^2 + y^2);
        if abs(rho) <= 0.84525
            sigma_xx = 2093*y/(pi*d^4/64);
            %tau = 100*(rho)/(pi*d^4/32);
            %VM(n) = sqrt(sigma_xx^2 + 3*tau^2);
            VM(n) = sqrt(sigma_xx^2);
            zz(n) = z;
            yy(n) = y;
        end
        n = n + 1;
    end
end
fig = scatter3(zz', yy', VM', [], VM', 'filled');
colormap(jet)
```

```
colorbar
view(0,90)
title('VM Stress Distribution at x = 10.9 in') %make sure title is correct!!!
ylabel('Height Above/Below NA (in)') %make sure units are correct!!!
xlabel('Distance Along z-axis (in)') %make sure units are correct!!!
hold on
x_line = [-1 1]; %make sure these values are correct!!!
y_line = [0 0];
p = plot(x_line,y_line,'k'); %this plots the Neutral Axis
p.LineWidth = 2;
```

Free Body Diagrams:



Final Table:

Feature	Final Design Parameters
"Length"	13.4

"min gear bore dia"	1.125
"max gear bore dia"	2.125
"min_d calculated"	1.6905
"2nd Iter Goodman F.S."	2.9096
"G2 Loc"	3.65
"G3 Loc"	8.95
"Teeth G2"	72
"Teeth G3"	16
"dp G2"	18
"dp G3"	4
"G2 FW"	4.375
"G3 FW"	5
"HP"	15
"Input RPM"	1750
"E (psi)"	2.9e+07
"S _{ut} (psi)"	2.95e+05
"S _e (psi)"	1.475e+05

BOM:

- 20 hp 256T electric motor 200V 1750 rpm 3 phase tefc 256TTFL14027: \$1413
- Gear spur 20 deg 72T gear: \$2485
 - TS472 Martin

- Gear Spur 20 Deg Steel Bs 16T gear: \$381
 - TS416
 - UPC: 697950263700
- External retaining ring black phosphate 1-1/8" OD(x50): \$9.12
- External retaining ring black phosphate 2" OD(x50): \$16.91
- Machine Key, 1008-1045 Carbon Steel, 1/8" x 1/8", 1" Long, Undersized(x10): \$3.23
- Machine Key, 1008-1045 Carbon Steel, 1/8" x 1/8", 3/4" Long, Undersized(x10): \$2.70
- Cylindrical Roller Bearings McMastercarr 25mm bearings(x2): \$140.62

Total = \$4451.58